Verify Numbers and Dates

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Verify Numbers and Dates before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image Generation Models

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Verify Numbers and Dates trace and record the first step you can explain without help.

Use W2: Trace the worked case for Claim support and citation evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Verify Numbers and Dates explanation against the supplied facts before discussing it.

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A draft claims18 of24 learners is**60%**, and gives an April31 deadline. Check both statements without inventing a deadline.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Verify Numbers and Dates target to a trace and a justified conclusion.

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Verify Numbers and Dates case and show the deciding evidence.

An answer says 15/20=**80%** and sets a deadline of February 30.

Verify both.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A report says 12 of 40 responses is 40%. **Recalculate** and state the correction with the denominator.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 16 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Verify Numbers and Dates adds to Verified Study Buddy and keep the evidence.

Route arithmetic to a calculator and dates to direct sources before presenting them.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Verify Numbers and Dates facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Verify Numbers and Dates", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Verify Numbers and Dates facts and keep your

> reasoning.

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Verify Numbers and Dates facts and keep your reasoning.

A report omits this qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Verify Numbers and Dates facts and keep your reasoning.

Five claims; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Verify Numbers and Dates facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Verify Numbers and Dates", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Verify Numbers and Dates

> A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts. Explain the labelled step before checking the changed condition.

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

Record: 15 of 20 fictional model trials passed. Draft: “**70%** passed; the meeting is on June 31.”

June has 30 days. No authoritative record gives the intended meeting date.

AI ACADEMY / AI-3/16

Verify Numbers and Dates

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Check denominator

There are 15 passes out of 20 total trials.



2. Calculate percentage

15 / 20 x 100=75%, not 70%.



3. Validate date

June 31 is invalid because June has 30 days.



4. Resolve missing intent

Ask the event owner for the correct date; do not assume June 30 or July 1.

CHANGE ONE THING
The count is revised to 14 passes out of 20, but no date record is added.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask the event owner for the correct date; do not assume June 30 or July 1.

1. Validate date

2. Check denominator

3. Resolve missing intent

4. Calculate percentage

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Resolve missing intent Ask the event owner for the correct date; do not assume June 30 or July 1. A percentage requires the correct numerator and denominator.

Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The count is revised to 14 passes out of 20, but no date record is added. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. **70%** now matches the count; June 31 remains invalid and unresolved.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **70%** now matches the count; June 31 remains invalid and unresolved. 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Repair a missing building block

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

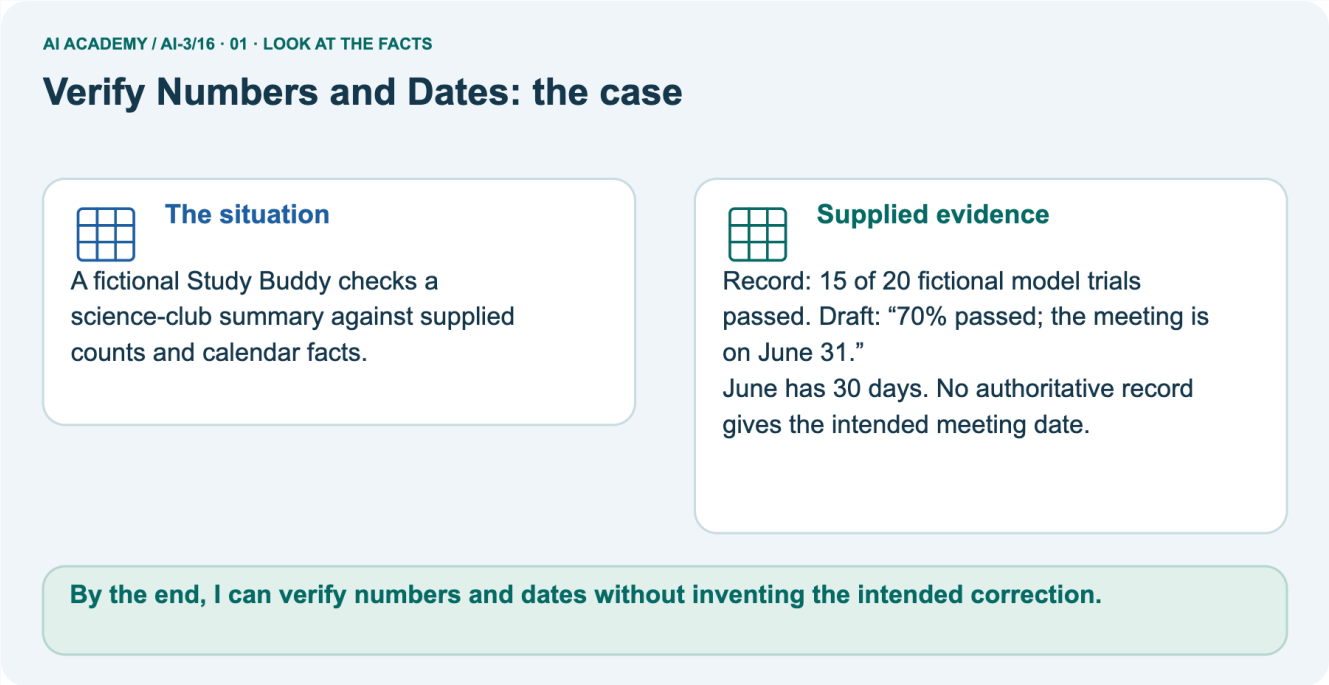
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Verify Numbers and Dates picture from its labels, then record a relevant limit.

01. Verify Numbers and Dates: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

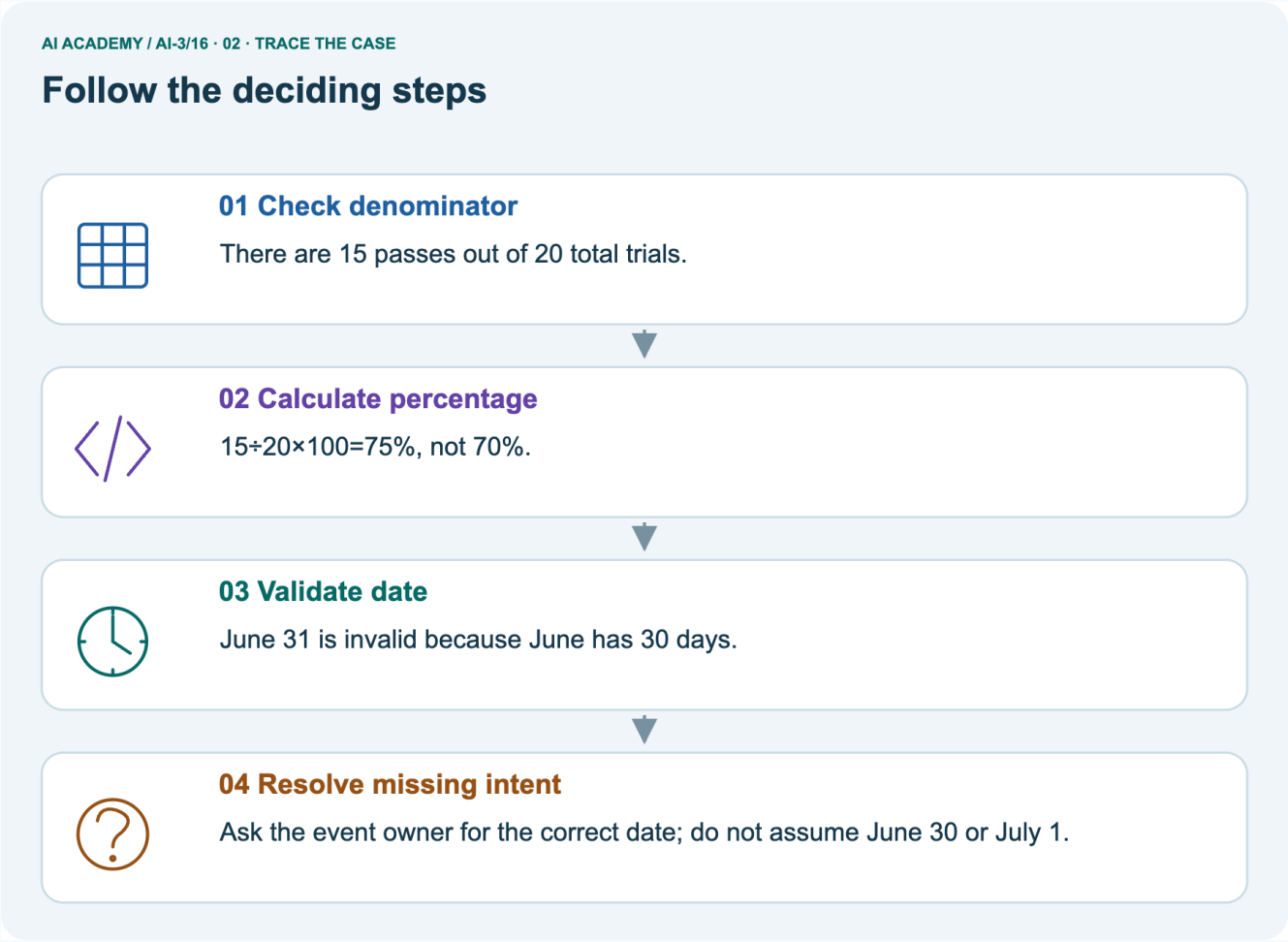
A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts. Supplied facts: Record: 15 of 20 fictional model trials passed.

Draft: “**70%** passed; the meeting is on June 31.” June has 30 days. No authoritative record gives the intended meeting date.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



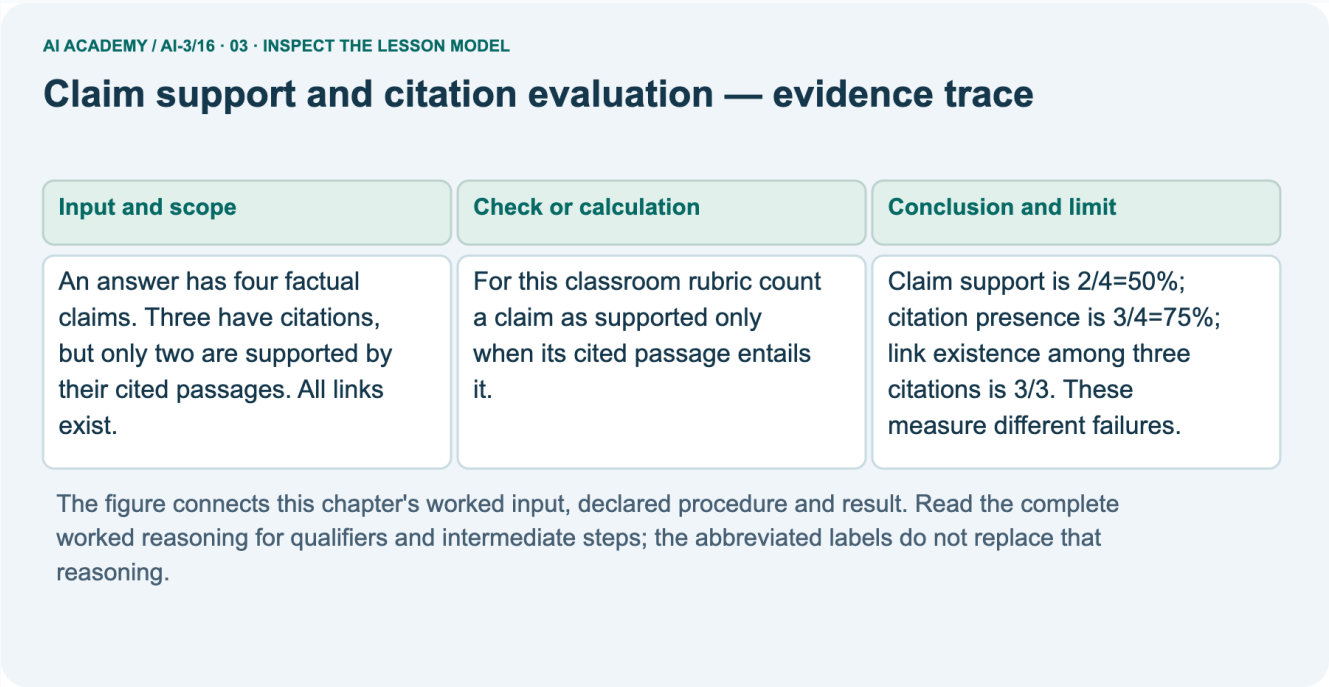
Follow the labelled picture with the matching explanation. Open the vector diagram

Check denominator: There are 15 passes out of 20 total trials. → Calculate percentage: 15÷20×100=75%, not **70%**. → Validate date: June 31 is invalid because June has 30 days. → Resolve missing intent: Ask the event owner for the correct date; do not assume June 30 or July 1.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Claim support and citation evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Claim support and citation evaluation — **evidence** trace. An answer has four factual claims. Three have citations, but only two are supported by their cited passages.

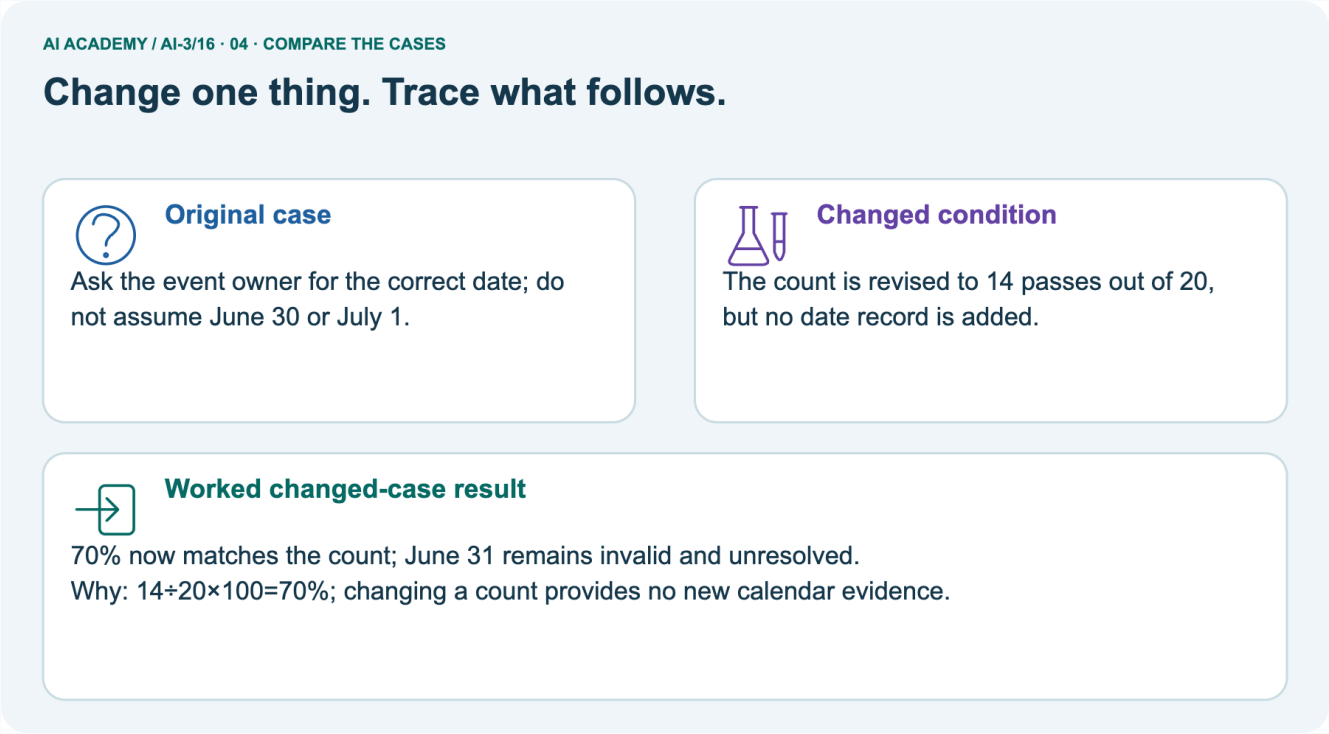
All links exist. | For this classroom rubric count a claim as supported only when its cited passage entails it. | Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3.

These measure different failures.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

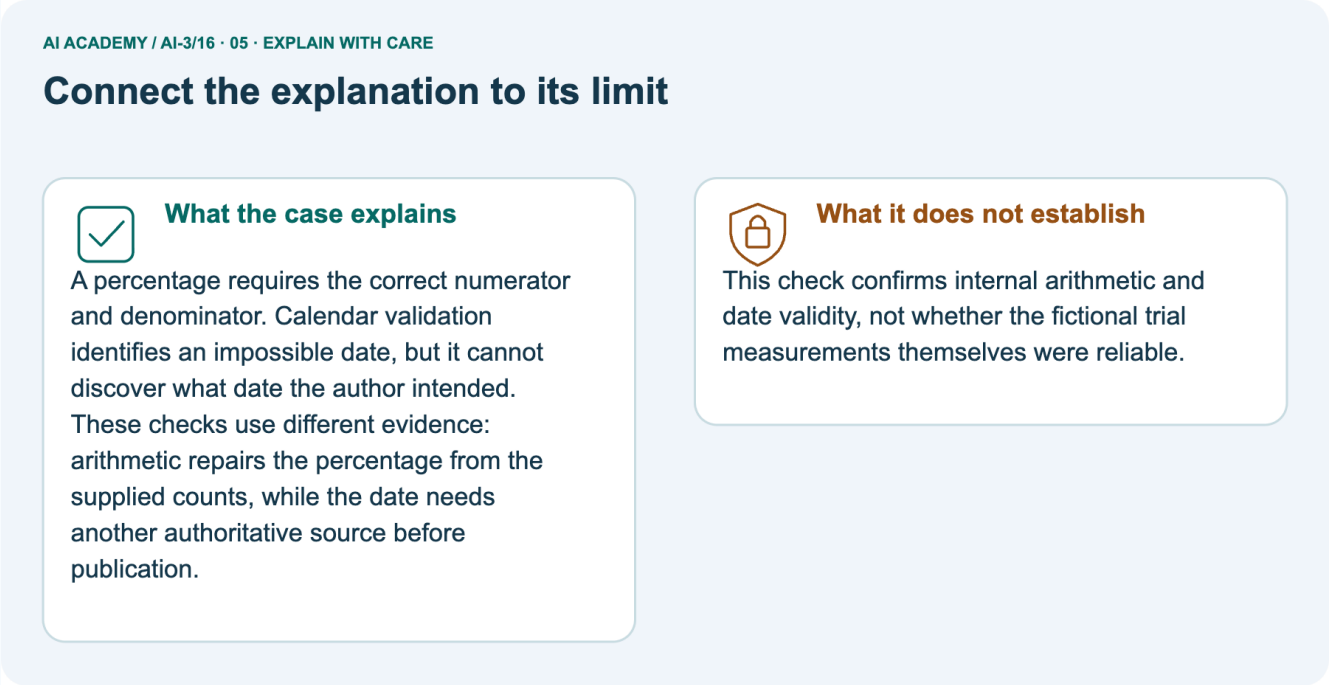
Original case: Ask the event owner for the correct date; do not assume June 30 or July 1. Changed condition: The count is revised to 14 passes out of 20, but no date record is added.

Expected result: **70%** now matches the count; June 31 remains invalid and unresolved. Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

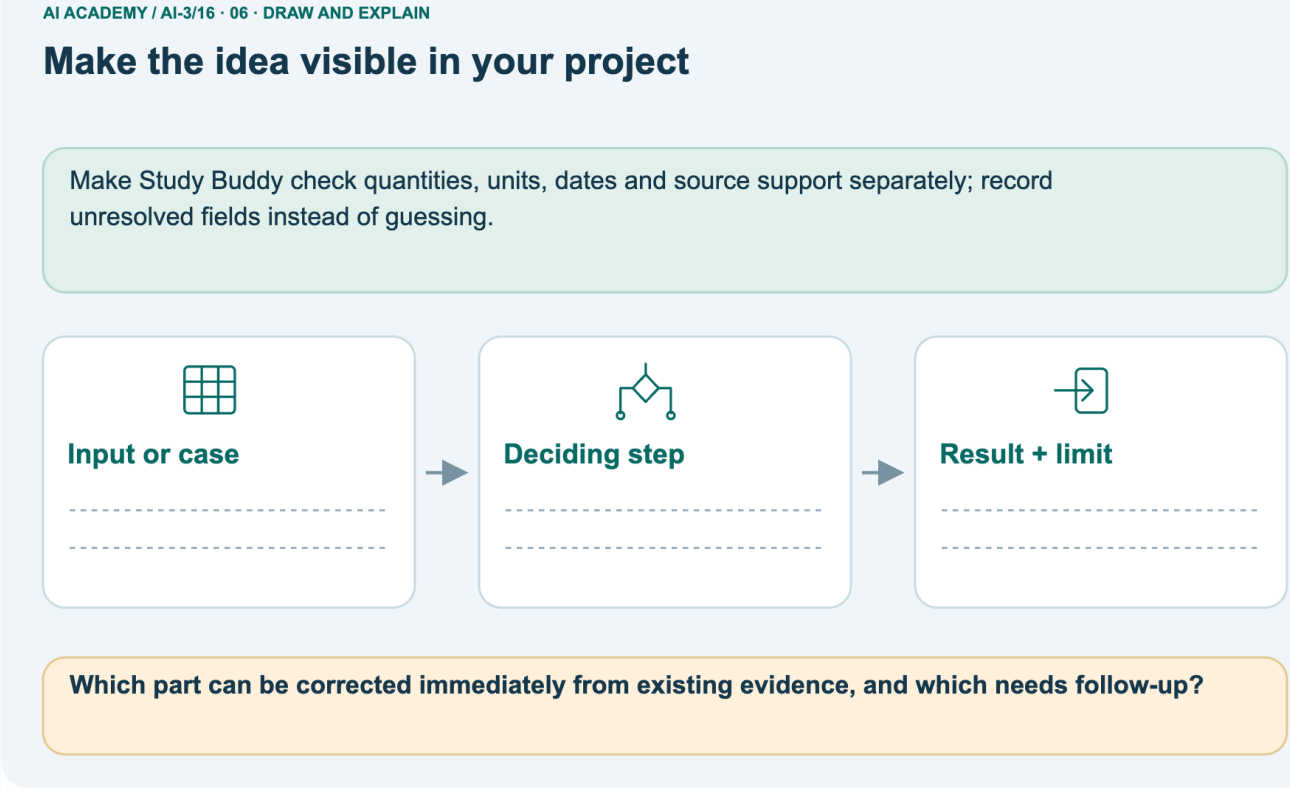
These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Limit: This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing. Draw your **input**, deciding step, result and limit.

Supported question: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.